PCT/PCM 11: The Restoration Bracket

4. Interpret the Figure

The correct PSF gives the best restoration result (23.1 dB PSNR), while mismatched PSFs cause ringing artifacts. In this figure, the too-long PSF performs worse (18.2 dB) than the too-short PSF (19.4 dB).

5. Super-Resolution Calculation

At ×2 super-resolution, the output has $2^2 = 4$ times as many pixels, so the directly observed fraction is 1/4 = 25%. At ×4, the output has $4^2 = 16$ times as many pixels, so the directly observed fraction is 1/16 = 6.25%.

6. Prediction

Yes. A Wiener-restored image can score worse than the blurred input if the assumed PSF is incorrect or the restoration amplifies noise and creates artifacts, increasing the reconstruction error.

Draft Degradation Card - ×4 Super-Resolution

Task: Reconstruct a high-resolution (HR) image from a low-resolution (LR) input.
Input/target: LR image as input; original HR image as target.
Forward model: LR = (HR * k) ↓4 + η, where k is a blur kernel, ↓4 means downsampling by a factor of 4, and η is optional noise.
Setup: Apply a fixed Gaussian blur kernel, downsample by 4 in each spatial dimension, and optionally add a small amount of noise. Keep the original HR image as ground truth.
Evaluation: Compare reconstruction with the HR target using PSNR and visual inspection. At ×4, the directly observed fraction is 1/16 (6.25%); the remaining detail must be inferred.